Stage -2
Implementation of
Human-In-The Loop Agent

Shoaib Khan

Introduction

Technologies Used

- Python
- LangChain
- ChromaDB
- HuggingFace Embeddings
- OpenRouter LLM
- JSON Storage
- pytest

Main Focus

- Multi-Agent Workflow
- Human-in-the-Loop Approval
- Reservation Lifecycle Management

Objectives

Main Goals

- Implement Human-in-the-Loop (HITL) workflow
- Create administrator approval agent
- Connect reservation workflow with approval system
- Maintain persistent reservation states
- Improve validation and security

Expected Outcome

- AI-assisted parking reservation system
- Administrator-controlled approval pipeline
- Multi-agent orchestration foundation

Multi-Agent Architecture

Agents Implemented

User Chatbot Agent

- Handles user interaction
- Answers parking-related questions
- Starts reservation workflow

Reservation Agent

- Collects reservation information
- Validates user inputs
- Maintains reservation state

Approval Agent

- Reviews reservations
- Approves or rejects requests
- Tracks reservation lifecycle

Reservation Workflow

Reservation Process

User starts reservation request

System collects:

- Name
- Verification ID
- Parking type
- Vehicle number
- Reservation dates

Reservation stored in pending queue
Admin reviews request

Reservation approved or rejected

Reservation Types

1. Standard Parking

2. VIP Parking

3. EV Charging Parking

Validation & Security

Security Features

- Prompt injection detection
- Sensitive data masking
- Identity validation
- Vehicle regex validation

Validation Rules

Aadhaar Validation

- 12-digit format

Vehicle Number Validation

Format: TS09AB1234

Date Validation

- No past dates
- End date after start date

Reservation Lifecycle Management

Reservation States

- Pending
- Approved
- Rejected

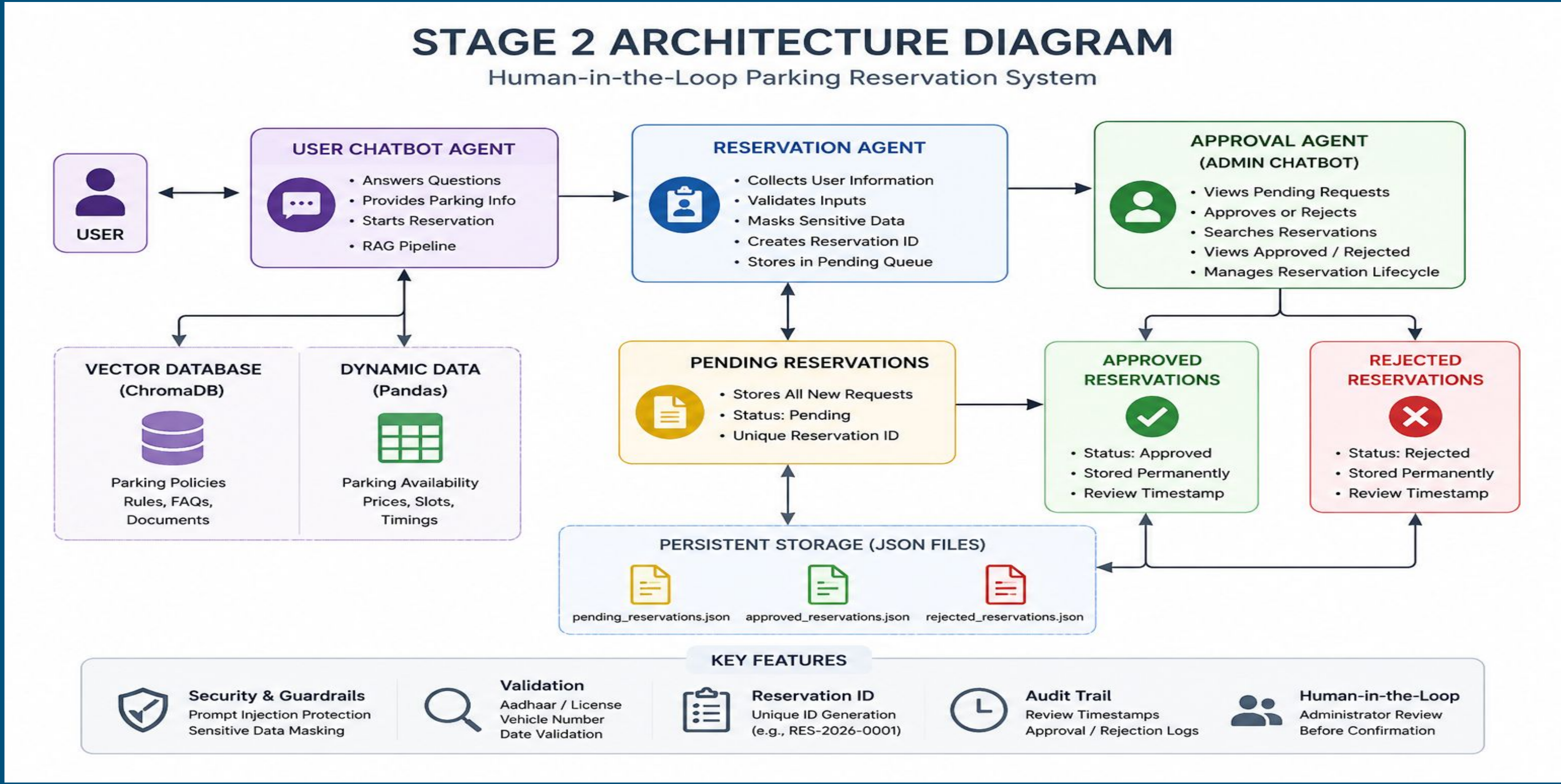
Reservation Tracking

Each reservation gets a unique ID:

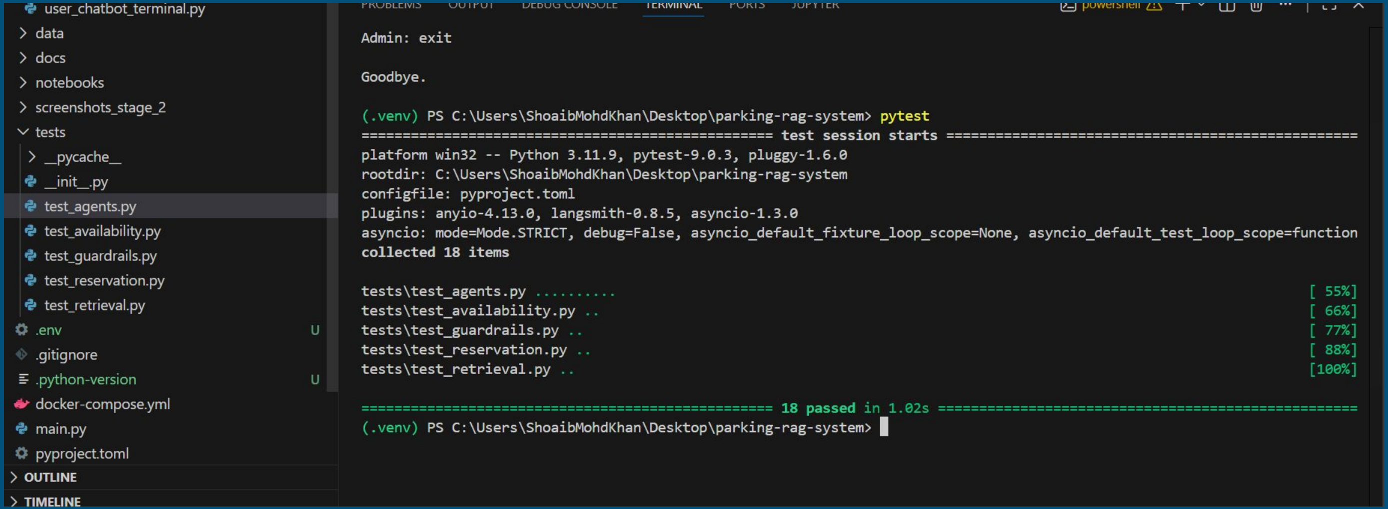
RES-2026-0001

Admin Commands

- list
- approve
- reject
- search
- view approved
- view rejected

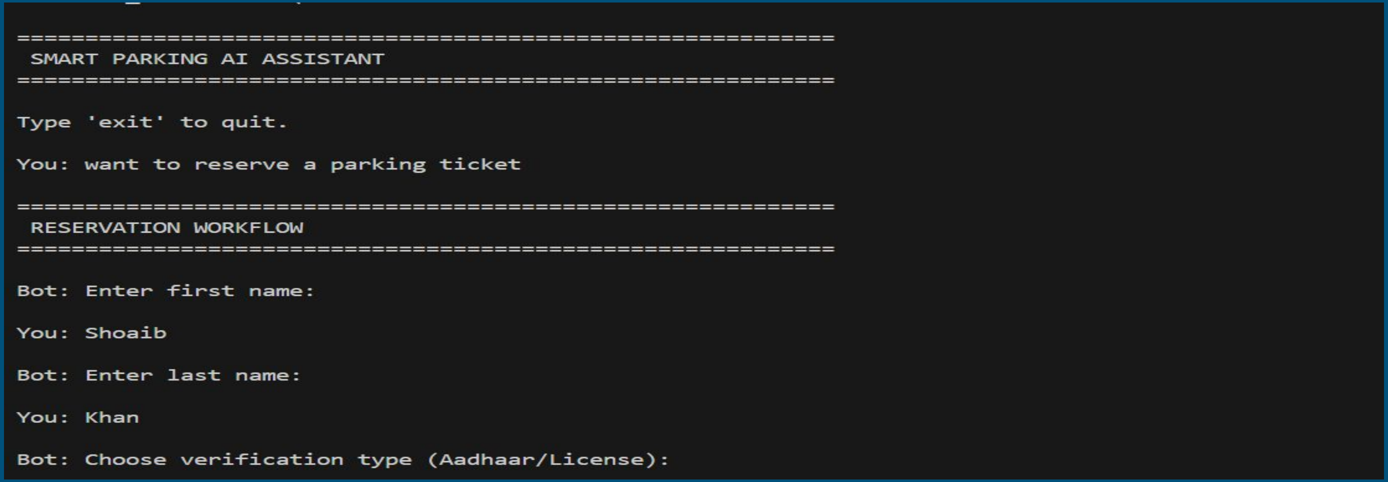

STAGE 2 ARCHITECTURE DIAGRAM
Human-in-the-Loop Parking Reservation System
USER
USER CHATBOT AGENT
• Answers Questions
• Provides Parking Info
• Starts Reservation
• RAG Pipeline
RESERVATION AGENT
• Collects User Information
• Validates Inputs
• Masks Sensitive Data
• Creates Reservation ID
• Stores in Pending Queue
APPROVAL AGENT (ADMIN CHATBOT)
• Views Pending Requests
• Approves or Rejects
• Searches Reservations
• Views Approved / Rejected
• Manages Reservation Lifecycle
VECTOR DATABASE (ChromaDB)
Parking Policies Rules, FAQs, Documents
DYNAMIC DATA (Pandas)
Parking Availability Prices, Slots, Timings
PENDING RESERVATIONS
• Stores All New Requests
• Status: Pending
• Unique Reservation ID
APPROVED RESERVATIONS
• Status: Approved
• Stored Permanently
• Review Timestamp
REJECTED RESERVATIONS
• Status: Rejected
• Stored Permanently
• Review Timestamp
PERSISTENT STORAGE (JSON FILES)
pending_reservations.json
approved_reservations.json
rejected_reservations.json
KEY FEATURES
Security & Guardrails
Prompt Injection Protection
Sensitive Data Masking
Validation
Aadhaar / License
Vehicle Number
Date Validation
Reservation ID
Unique ID Generation
(e.g., RES-2026-0001)
Audit Trail
Review Timestamps
Approval / Rejection Logs
Human-in-the-Loop
Administrator Review
Before Confirmation

Screenshots



```
user_chatbot_terminal.py
> data
> docs
> notebooks
> screenshots_stage_2
v tests
  > __pycache__
  __init__.py
  test_agents.py
  test_availability.py
  test_guardrails.py
  test_reservation.py
  test_retrieval.py
.env                U
.gitignore
.python-version     U
docker-compose.yml
main.py
pyproject.toml
> OUTLINE
> TIMELINE
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  JUPYTER          powershell

Admin: exit

Goodbye.

(.venv) PS C:\Users\ShoaibMohdKhan\Desktop\parking-rag-system> pytest
============================================= test session starts ==============================================
platform win32 -- Python 3.11.9, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\ShoaibMohdKhan\Desktop\parking-rag-system
configfile: pyproject.toml
plugins: anyio-4.13.0, langsmith-0.8.5, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 18 items

tests\test_agents.py ..........                                                                          [ 55%]
tests\test_availability.py ..                                                                            [ 66%]
tests\test_guardrails.py ..                                                                              [ 77%]
tests\test_reservation.py ..                                                                             [ 88%]
tests\test_retrieval.py ..                                                                               [100%]

============================================== 18 passed in 1.02s ==============================================
(.venv) PS C:\Users\ShoaibMohdKhan\Desktop\parking-rag-system>
```



```
============================================================
 SMART PARKING AI ASSISTANT
============================================================

Type 'exit' to quit.

You: want to reserve a parking ticket

============================================================
 RESERVATION WORKFLOW
============================================================

Bot: Enter first name:

You: Shoaib

Bot: Enter last name:

You: Khan

Bot: Choose verification type (Aadhaar/License):
```

```
Bot: Choose verification type (Aadhaar/License):

You: Aadhaar

Bot: Enter verification ID:

You: 123456789000

Bot: Choose parking type (Standard/VIP/EV):

You: standard

Bot: Enter car number (Format: TS09AB1234):

You: RR09RR0000

Bot: Enter reservation start date (YYYY-MM-DD):

You: 2026-09-09

Bot: Enter reservation end date (YYYY-MM-DD):

You: 2026-09-18

Bot: Reservation details completed.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

```
============================================================
 ADMIN APPROVAL AGENT
============================================================

Commands:

list
approve RES-2026-0001
reject RES-2026-0001
search RES-2026-0001
view approved
view rejected
exit


Admin: approve RES-2026-0001

Reservation RES-2026-0001 approved.


Admin: view approved

============================================================
Reservation ID: RES-2026-0001
============================================================
first_name: Shoaib
last_name: Khan
verification_type: Aadhaar
verification_id: XXXXXXXX9000
parking_type: standard
car_number: RR09RR0000
start_date: 2026-09-09
end_date: 2026-09-18
status: approved
reservation_id: RES-2026-0001
created_at: 2026-05-28 11:57:18.312446
reviewed_at: 2026-05-28 11:57:34.904112
```

Thank You